

VLSI Internship – Task 2 Report

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1 Introduction

Verilog HDL (Hardware Description Language) is widely used in digital electronics and VLSI design for modeling, simulation, and verification of digital systems. It allows designers to describe hardware behavior using programming constructs and verify the functionality of circuits through simulation.

In this task, basic logic gates, Half Adder, and Full Adder circuits were implemented using Verilog HDL. The circuits were simulated using EDA Playground with the Icarus Verilog simulator, and the outputs were verified using waveform analysis.

2 Basic Logic Gates

2.1 AND Gate

The AND gate produces an output of 1 only when both inputs are 1.

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Verilog Code:

```
module and_gate(  
    input A,  
    input B,  
    output Y  
);  
  
assign Y = A & B;  
  
endmodule
```

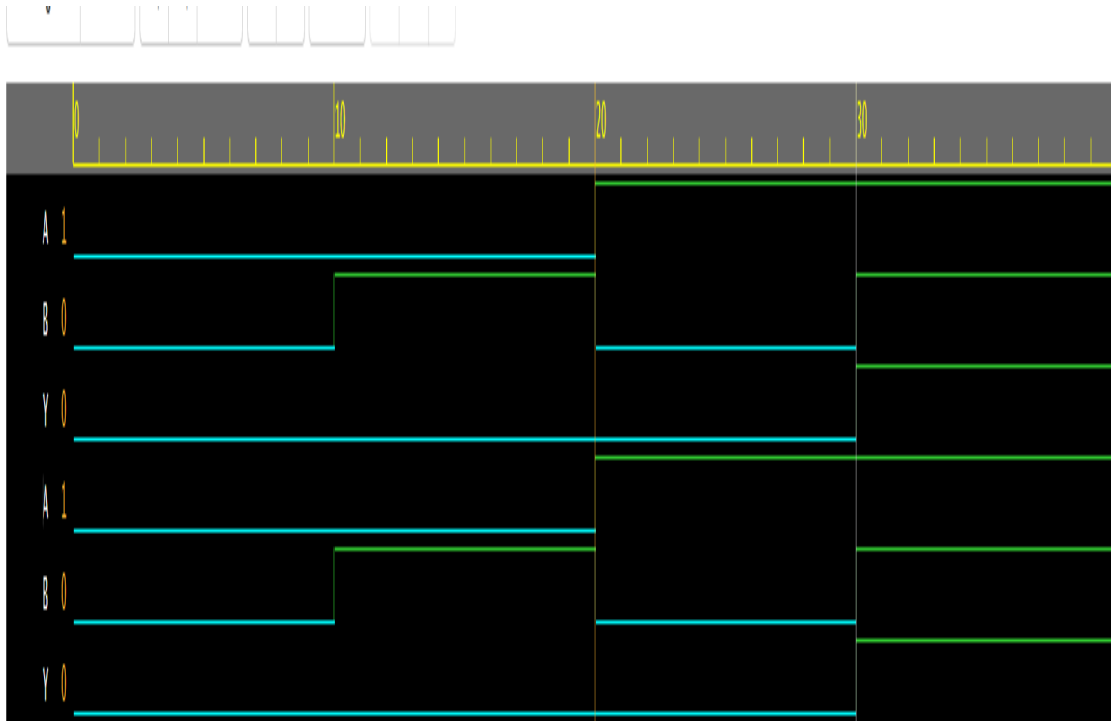


Figure 1: AND Gate Waveform

2.2 OR Gate

The OR gate produces an output of 1 when at least one input is 1.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Verilog Code:

```
module or_gate(
    input A,
    input B,
    output Y
);

assign Y = A | B;

endmodule
```

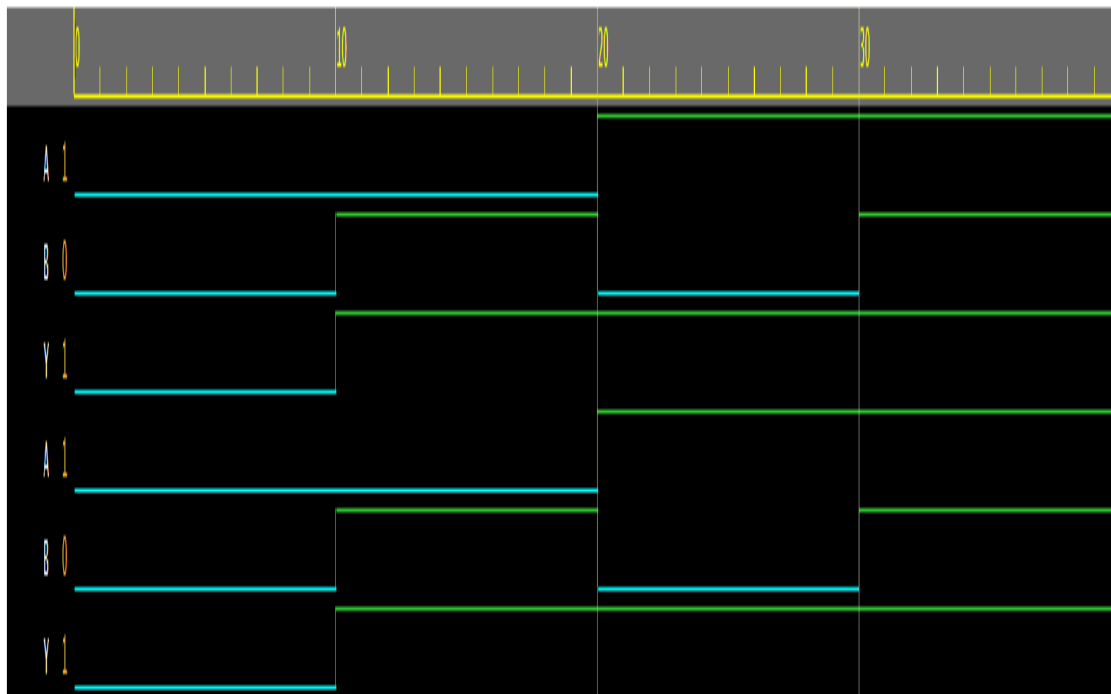


Figure 2: OR Gate Waveform

2.3 NOT Gate

The NOT gate produces the inverse of the input.

A	Y
0	1
1	0

Verilog Code:

```

module not_gate(
    input A,
    output Y
);

assign Y = ~A;

endmodule

```

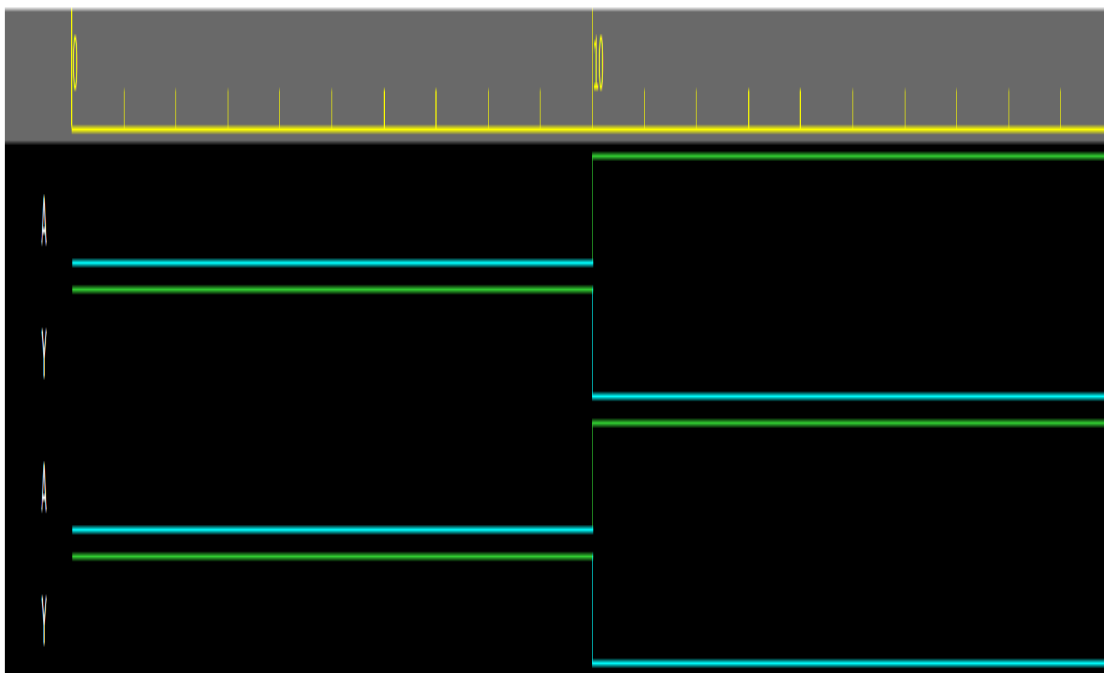


Figure 3: NOT Gate Waveform

2.4 NAND Gate

The NAND gate produces the inverse of the AND gate output.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Verilog Code:

```
module nand_gate(
    input A,
    input B,
    output Y
);

assign Y = ~(A & B);

endmodule
```

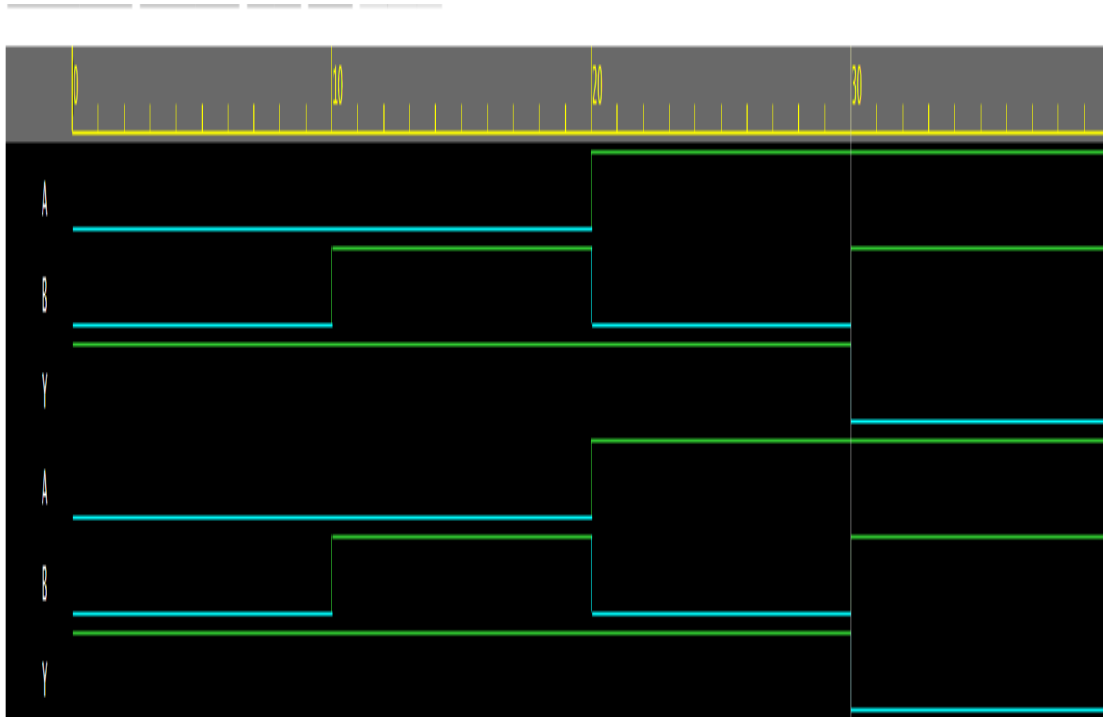


Figure 4: NAND Gate Waveform

2.5 NOR Gate

The NOR gate produces the inverse of the OR gate output.

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Verilog Code:

```
module nor_gate(
    input A,
    input B,
    output Y
);

assign Y = ~(A | B);

endmodule
```

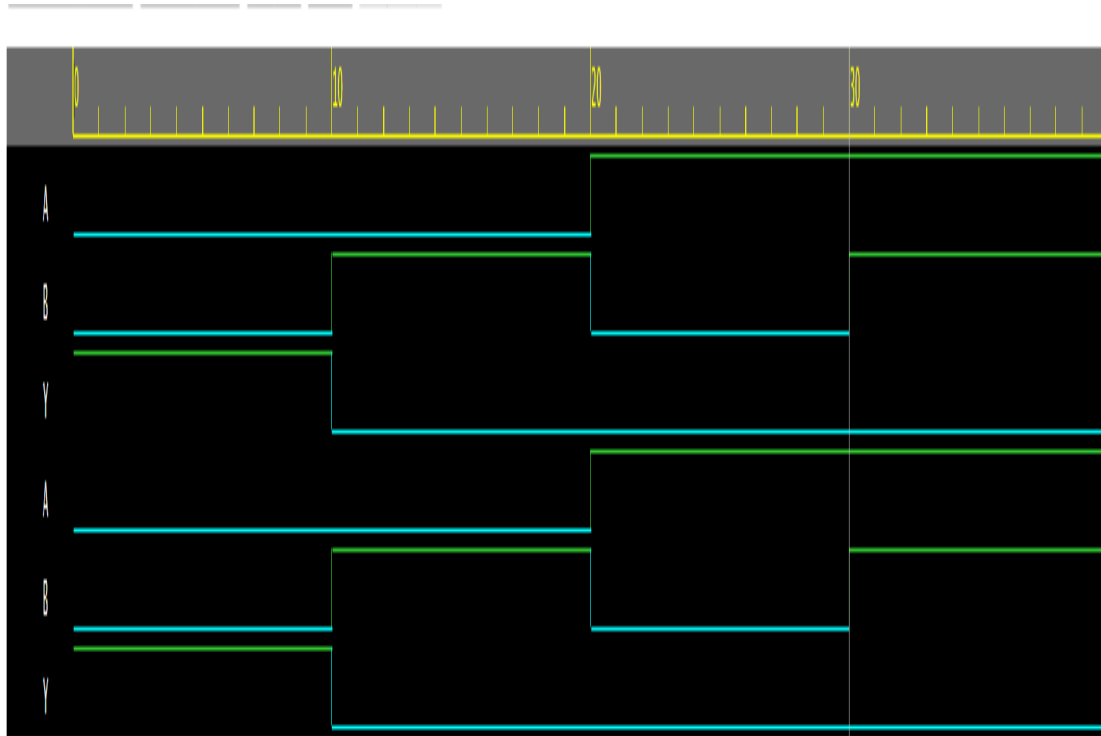


Figure 5: NOR Gate Waveform

2.6 XOR Gate

The XOR gate produces an output of 1 when the inputs are different.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Verilog Code:

```
module xor_gate(
    input A,
    input B,
    output Y
);

assign Y = A ^ B;

endmodule
```

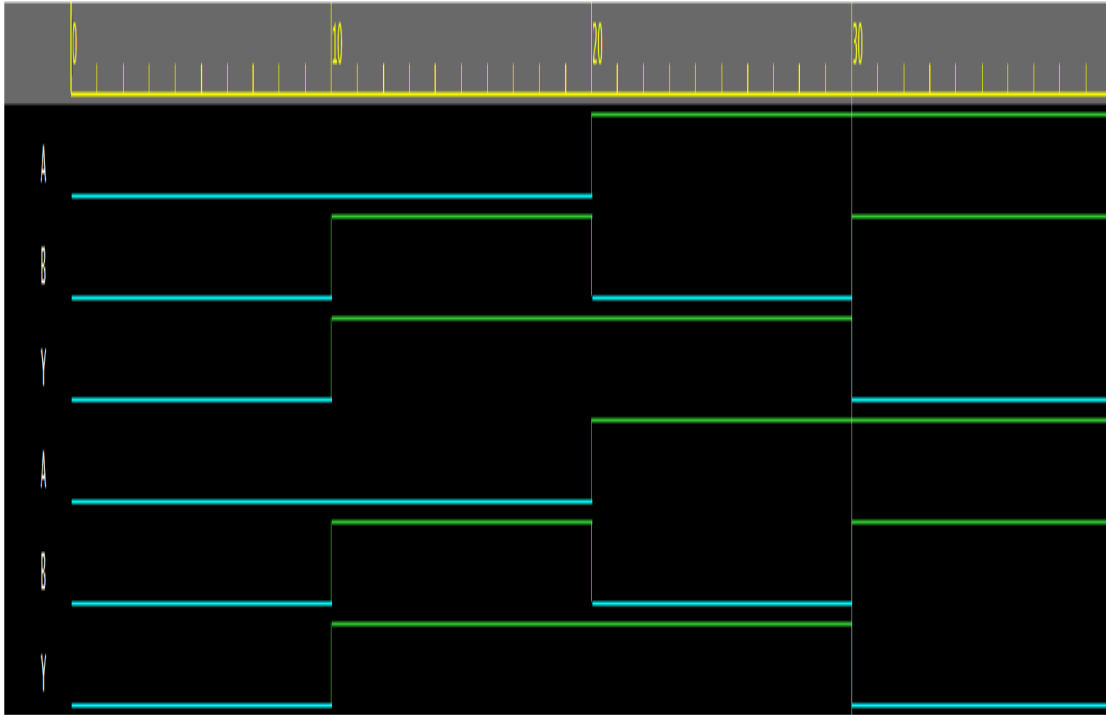


Figure 6: XOR Gate Waveform

3 Half Adder

A Half Adder is a combinational circuit used to add two binary bits.

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = A \cdot B$$

3.1 Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Verilog Code:

```
module half_adder(
    input A,
    input B,
    output Sum,
```

```

        output Carry
    );

    assign Sum = A ^ B;
    assign Carry = A & B;

endmodule

```

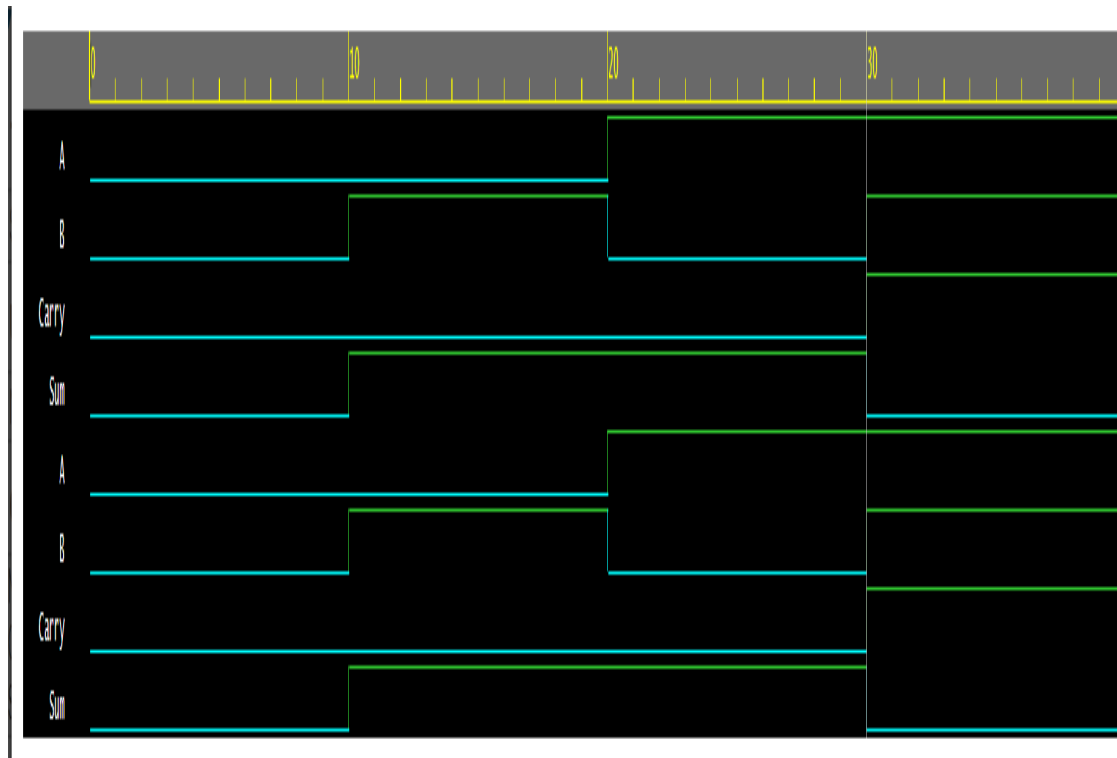


Figure 7: Half Adder Waveform

4 Full Adder

A Full Adder is a combinational circuit used to add three binary inputs including carry input.

$$\text{Sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = AB + BCin + ACin$$

4.1 Truth Table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Verilog Code:

```
module full_adder(  
    input A,  
    input B,  
    input Cin,  
    output Sum,  
    output Cout  
);  
  
assign Sum = A ^ B ^ Cin;  
assign Cout = (A & B) | (B & Cin) | (A & Cin);  
  
endmodule
```

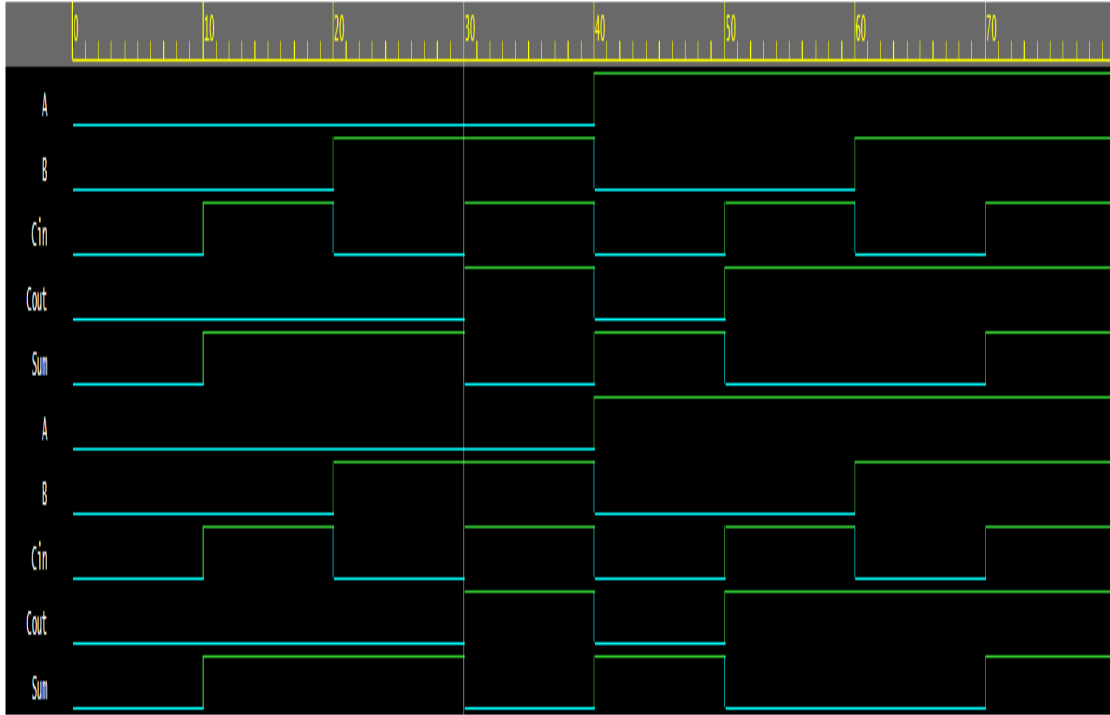


Figure 8: Full Adder Waveform

5 Simulation

The Verilog modules were simulated using EDA Playground with the Icarus Verilog simulator. Testbenches were written for each circuit to apply different input combinations and verify the outputs. Waveforms generated during simulation confirmed the correctness of the logic gate operations, Half Adder, and Full Adder circuits.

6 Conclusion

In this task, various basic logic gates along with Half Adder and Full Adder circuits were successfully implemented using Verilog HDL. The designed circuits were verified through simulation and waveform analysis. This task provided practical understanding of hardware description languages, digital circuit simulation, and verification techniques used in VLSI design.